Battery Storage Unit

Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Battery Storage Unit (Lithium-ion)

What it is used for

A lithium-ion battery storage unit stores electrical energy and supplies power to connected equipment or loads through a battery management system and associated power electronics. Operate it within the manufacturer's voltage, temperature, current, and charging limits, and monitor alarms to reduce the risk of electric shock, fire, and thermal runaway.

Main parts

- Lithium-ion battery modules or cells
- Battery management system (BMS)
- Battery enclosure and protective covers
- Power conversion equipment or inverter, if fitted
- Electrical isolators, fuses, and circuit breakers
- Cooling or ventilation system and monitoring interface

Before you start

1. Confirm the unit's chemistry, rated voltage, capacity, operating limits, and approved application.
2. Check the enclosure, cables, terminals, connectors, isolators, and visible modules for damage, swelling, corrosion, or leakage.
3. Review the BMS display for state of charge, temperature, voltage, alarms, and fault history.
4. Verify that earthing, overcurrent protection, emergency isolation, and installation clearances meet the approved design.
5. Confirm that cooling, ventilation, fire detection, and other required protective systems are operational.
6. Ensure that only authorised personnel are present and that the emergency response procedure is available.

How to use

1. Inspect the installation and confirm that no active alarm or unresolved safety fault is present.
2. Verify that the connected load, inverter, and charging source are compatible with the battery unit.
3. Confirm that all covers are closed and protective interlocks are in their required operating state.
4. Energise the system using the manufacturer's prescribed startup sequence and approved isolator procedure.
5. Check the BMS and power conversion interface for normal voltage, current, temperature, and communication status.

6. Start charging or discharging only within the approved operating limits and site energy-management procedure.
7. Monitor cell or module temperatures, state of charge, current, and alarms during operation.
8. If a warning occurs, follow the approved response procedure and reduce or stop operation when required.
9. If smoke, unusual odour, hissing, swelling, rapid temperature rise, or fire is observed, keep clear, raise the alarm, and follow the emergency plan; do not open the enclosure.
10. For shutdown, stop the load or charging process and isolate the system using the approved sequence.
11. Allow the system to reach the required safe state before any inspection or maintenance; use the specified electrical isolation and verification procedure.
12. Record operating data, alarms, inspections, and maintenance actions in the site log.

⚠ Safety

- Only qualified and authorised personnel should install, commission, isolate, or service the battery unit.
- Battery terminals and internal circuits may remain electrically hazardous even when the inverter is off; follow the approved isolation procedure.
- Never open battery modules, bypass the BMS, short-circuit terminals, or use unapproved chargers.
- Do not operate a battery that is damaged, swollen, leaking, overheated, or showing a critical fault.
- Keep combustible materials clear and maintain required fire detection, ventilation, and emergency access.
- If thermal runaway or fire is suspected, evacuate the affected area and contact emergency responders; follow the site-specific battery fire plan.

Care & storage

- Inspect the enclosure, terminals, cables, cooling equipment, and protective devices according to the maintenance schedule.
- Review BMS alarms and trends in cell voltage, temperature, and capacity for abnormal changes.
- Keep air inlets and cooling paths clear; do not obstruct ventilation openings.
- Arrange qualified testing and servicing of protection devices, connections, and power electronics.
- Follow the manufacturer's requirements for storage state of charge, temperature, and long-term shutdown.

Common mistakes to avoid

- Operating outside the specified voltage, current, or temperature limits.
- Ignoring BMS alarms, unusual heating, swelling, or abnormal odours.
- Opening the enclosure or servicing electrical parts without proper isolation and authorisation.
- Blocking cooling paths or using incompatible charging equipment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.